

Blue Bike contribution to transit connectivity and greenhouse gas emission reduction in the Boston Metropolitan Area

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Abstract

Blue Bike has been a crucial component of sustainable urban mobility in the Boston Metropolitan Area. However, there is fragmented evidence of its contribution to transit connectivity and greenhouse gas emission reduction. Therefore, this study set out to synthesize evidence on Blue Bike's contribution to transit connectivity and greenhouse gas emissions reduction in the Boston Metropolitan Area. The study employed a systematic review and a meta-synthesis approach drawing on peer-reviewed and high-quality grey literature identified through ScienceDirect and screened using defined exclusion and inclusion criteria. The quality of the selected studies included was evaluated using the Joanna Briggs Institute Critical Appraisal Checklist. The findings indicate that Blue Bike is closely integrated with Boston's MBTA transit nodes, enhancing first-and last-mile connectivity as well as

supporting multimodal travel. It was found that Blue Bike primarily substitutes short-distance vehicular (car) trips and is predominantly used for routine urban mobility. In addition, Blue Bike contributes to lower transportation related emission through reduced reliance on private vehicles. This emphasizes the relevance of integrating bike-sharing and public transit systems to foster sustainable urban mobility and inform transport planning and climate policy.

Keyword

Boston, Greenhouse Gas Reduction, Transit Connectivity, Urban Mobility, Bikeshare Sustainability.

1.0 Introduction

Urban transport systems have been one of the major contributors to greenhouse gas emissions, reported by several studies. For instance, a study by Wang et al (2021) revealed that globally, motorized urban transport systems contribute about 24% of carbon dioxide (CO₂) emissions within urban areas or cities in the year 2021. Kou et al (2020) further posits that in recent times, there has been an increasing attention shifting toward shared and active urban transport modes, specifically bike sharing as a strategic initiative that tends to offer social and environmental benefits as cities strive toward ensuring sustainable mobility. Bike sharing initiatives have gained much attention due to their potential contribution to greenhouse gas emission reduction and provision of flexible alternative modes for urban transport, making it easier for first and last mile connections from urban households to various public transport systems. Prior studies (Elliot Fishman, Paul Schepers, 2015; Shaheen Susan, 2019) have reported that the integration of bike sharing into the urban transport system has the potential to reduce greenhouse gas emissions. Because replacing a non-motorized transport system with a motorized system tends to reduce the emissions released into the urban atmosphere by vehicular motors.

A study by Shaheen (2019) revealed that European cities, including Barcelona and Paris, witnessed a reduction in carbon emissions of 5-10% as a result of urban households or city residents patronizing bike sharing for short trips. Hu & Creutzig (2022) affirmed the contributing role of bike share systems on carbon dioxide mitigation within cities in China. Within the context of China, Hu & Creutzig (2022) indicated that millions of short trips or mobility within urban centers and cities using private or commercial vehicles have been replaced by bike sharing systems, contributing to a notable decline of CO₂ emissions. This has highlighted the relevance of bike sharing as a crucial strategy to reduce the high dependency rate of the motorized transport system. This represents an attempt toward achieving a sustainable urban transport system and avoiding over-reliance on the fossil fuel mode of transportation. Within the context of the United States of America, studies such as (Ding et al., 2025; Fraser et al., 2025; Kou et al., 2020a) have presented empirical evidence ascertaining the effectiveness of bike share systems in enhancing urban transport and carbon emission reduction, especially in dense and multimodal cities. Kou et al (2020) reported that the integration of bike share trips within cities in the US has led to a net reduction in greenhouse gas emissions to approximately 8kg CO₂ per trip. Ding et al (2025) further recount that the introduction of bike share to complement transit networks within cities in the United States of America has provided alternatives, encouraging modal substitution for city residents.

Empirical research in the USA by Fraser et al (2025) affirmed these findings from prior studies on the relevance of bike share in addressing urban transport and environmental issues. According to Fraser et al (2025), bike share terminals situated close to transit nodes, particularly those within the catchment area of 400-800 meters close to public transit, have improved accessibility to urban transport and easy mobility and have lessened the need for capital-intensive infrastructure expansion. Moreover, it has also contributed to urban transport resilience and health outcomes by ensuring active transport and the reduction of congestion.

Several studies that have explored bike share initiatives and urban transportation have focused on trips and mobility patterns using aggregated trip data, with limited exploration of

their contribution and linkage to localized emission outcomes. This makes Boston suitable to explore bike share contributions and impact on localized greenhouse gas emission outcomes in the US. Through Boston's comprehensive climate action targets proposed through the Carbon-Free Boston Plan, Boston serves as one of the cities within the United States committed to achieving carbon neutrality by the year 2050 and a substantial reduction in transport-related emissions (Ding et al., 2025). According to Cleveland et al (2019) and Ding et al (2025), despite the policy comment, Boston's transport sector remains the largest contributor to greenhouse gas emissions, accounting for about 40% of total emissions.

As an attempt to curb these environmental detriments, Boston Metropolis has invested in blue bike systems to ensure a more sustainable urban transportation system. This seeks to enhance urban mobility and reduce the greenhouse gas emissions rate (Ethier et al., 2024). This initiative was launched in 2011, operating across cities such as Boston, Cambridge, Somerville, and Brookline to provide an alternative mode of urban transport for city residents (Ethier et al., 2024).

Through this initiative, more than 4000 bicycles together with 450 docking stations have been deployed, with about 70% located within a sphere of 500 meters of Massachusetts Bay Transportation Authority (MBTA) rail or bus stops (James et al., 2021; Swindell, 2024). These systems are positioned strategically to enable and ensure ease first and last-mile connectivity to urban transit networks and short trips as a substitute to reduce over-dependency on car trips (Swindell, 2024). The improvement in Boston's urban transport system resulted in the extension of the Blue Bike transport mode to low-income and underserved communities, fostering mobility equity, which aligned with the Carbon Free Boston Plan initiative (James et al, 2021). Despite the benefits provided by the Blue Bike initiative to Boston's urban transport and environmental sustainability agenda, there is unclear and fragmented evidence in the literature regarding how the Blue Bike initiative has contributed to greenhouse gas emission reduction. For instance, previous studies on micromobility and bike share within the context of Boston have primarily focused on different dimensions of system performance. To mention a few, Boeri (2025) studied micromobility user behaviour and vehicular usage patterns, Smith

(2022) examined the effects of policy interventions and utilization of bike-share. Again, Rush et al (2022) analyzed shared mobility data availability and usage trends, while Joyce-Johnson (2025) also studied the interaction between shared micromobility and public transit in enhancing car-free living. Ding et al (2025) assessed the environmental benefits of Boston's bike-sharing system. However, the study primarily concentrated on the broader environmental outcomes and not solely on its contributions to greenhouse gas emission reduction. Therefore, evidence on the subject matter remains fragmented with little attempt to explicitly synthesize how Blue Bike contributes to transit connectivity and greenhouse gas emission reduction.

Therefore, this study set out to address this gap in existing knowledge by providing a synthesis and an integrated assessment of Blue Bike's contribution to greenhouse gas emission reduction within the Boston Metropolitan Area. This study seeks to achieve this by answering the following research questions: a) What is the spatial linkage between Blue Bike stations and MBTA transit nodes? b) To what extent does Blue Bike substitute vehicular trips, trip characteristics, usage intensity, and threshold, and c) What has been the contribution of the Blue Bike initiative in reducing Greenhouse Gas?

2.0 Literature Review

2.1 The concept of Bike-sharing System.

Bike-sharing systems, particularly Blue-Bike, have evolved rapidly, especially within cities in developed countries as an alternative mode to promote sustainable urban mobility, providing low-cost, environmentally friendly, and flexible urban transport alternative to city residents. From the early 2000s, the system has been rapidly expanded and evolved from mere docked schemes to advanced and technological dockless systems with hybrid models enhancing urban mobility and accessibility (Todd et al 2021, Guo et al 2022). According to Guo (2022), bike-share networks are often located strategically to enhance first and last-mile connectivity, purposely to link transit nodes, commercial centres, working areas, and residential areas. Todd et al (2021) also revealed that the performance, travel behaviour, and accessibility of

bike-share systems are often influenced by the spatial arrangement of docking stations and the availability of bicycles.

Studies such as Mahajan & Saches (2024) have demonstrated that bike-share utilization is also influenced by the proximity of the system to high-density and mixed-use areas and public transport nodes. However, studies like Barker et al (2024) have raised concerns with regard to equity, indicating that these system and associated infrastructure is mostly concentrated in neighbourhoods that are economically advantaged compared to low-income and peripheral communities, leading to spatial inequity. Bike-sharing systems have played a crucial role as a substitute for conventional transport nodes, especially short vehicular trips within cities, which are primarily rendered by taxi or ride-hailing services. Studies have shown evidence of the contribution of bike-sharing across cities in Europe, Asia, and North America, highlighting how the system has encouraged modal shift by replacing vehicular trips and short-distance trips (Todd et al 2021, Guo et al 2022).

For instance, in Europe, studies such as (Creutzig et al., 2022; Geels, 2018; Ricci, 2015; Shaheen Susan, 2019) have presented empirical evidence demonstrating how the integration of bike sharing has contributed to greenhouse gas emissions in cities like Amsterdam and Copenhagen. Findings from Creutzig et al (2022) revealed that the introduction of cycling as an alternative mode for urban mobility and transport contributed to about a 15-25% reduction in greenhouse emissions in Amsterdam and Copenhagen for the past two decades. Again, in Paris, Gu et al (2019) ascertained that the Velib initiative, which was executed in 2007, contributed to approximately 9% modal shift among city residents and urban households, reducing over-reliance on car-trips to cycling between the years 2012 and 2020.

Within the context of Asia and North America, studies such as (Faghih-imani & Eluru, 2016; Kou et al., 2020a; Mohamed & Mahmoud, 2015) have presented similar empirical evidence. For example, in China, the Shanghai Mobike and Beijing Public Bike-share have been an alternative mode of urban transport, leading to a transition from over-reliance on vehicles (Kou et al., 2020a; Ma et al., 2023). Also, the introduction of bike-sharing has not only provided easy urban transport options to city residents in Shanghai and Beijing (Kou et al, 2020) but has

contributed to an estimated reduction of 40,000 tons of CO₂ emissions annually as a result of being a substitute for car trips. The integration of the Citi Bike network in New York City as a complement to the Metropolitan Transit Authority (MTA) resulted in about 10 million car trips being replaced by bike and cycling in the period of 2017 to 2023, leading to a reduction of 3,000 tons of CO₂ emissions (Boeri, 2025). However, the intensity of usage of the system varies across cities (Guo 2020). This is influenced by factors such as characteristics of the built environment, population density, weather conditions, and availability of cycling infrastructure. In terms of user characteristics, literature has revealed that typical users comprise young adults, working professionals, students, and commuters who value convenience, affordability, and environmental sustainability.

2.2 Overview of Blue Bike in the Boston Metropolitan Area.

Within the context of Boston Metropolis, studies have asserted that Blue Bike plays a crucial role in providing transit connectivity, a substitute for car trips, shorter distances, and urban mobility alternatives to city residents, enhancing the metropolis' multimodal transport ecosystem. The Blue Bike system, also known as the Hubway, was launched in the Boston Metropolitan Area in July 2011, a collaborative urban transport intervention put forward by the Metropolitan Area Planning Council (MAPC), the City of Boston, and the Operator Alta Bicycle Share (MAPC 2023). According to the City of Boston's Report, the Operator Alta Bicycle Share system was rebranded as the Blue Bike following corporation and sponsorship from the Blue Cross Blue Shield of the City of Massachusetts in May 2018 (City of Boston, 2018).

The rebranding and the cooperation sponsorship led to the expansion of the system across parts of Boston and other cities, including Cambridge, Everett, Brookline, and Somerville, increasing fleet size and the number of docking stations. Again, MAPC (2023) records revealed that over 4,000 bikes and 450 stations are operating within Boston Metropolis and sister cities, serving approximately fourteen million trips. The Blue Bike system is a strategy initiative adopted as a blueprint for the Go Boston mobility plan for Agenda 2030 with the sole objective

of improving multimodal accessibility, enhancing active transportation, and reducing greenhouse gases emitted through the urban transport system (City of Boston 2023).

According to Fraser et al (2025), the majority of Blue Bike trips within Boston and sister cities begin and end within a buffer of 500 meters of stop points of public transit, emphasizing the role of Blue Bike in enhancing first and last-mile trips of urban residents. The patronage of the system is reported to have grown, and it is fueled by factors such as network expansion, seasonal variation, and improved digital integration. According to the NACTO (2023) report, between the period of 2018 and 2023, about 25% increase in annual ridership was recorded, especially around summer and early fall. Studies such as (Basu & Ferreira, 2021; Ethier et al., 2024) have revealed that users who patronized Blue Bike services are primarily urban commuters within the age bracket 25 to 44, with concentration in proximity to corridors of major transit. Also, the introduction of the Bike for All, a discounted initiative targeting equity goals and membership, also enhanced the participation of low-income groups and women.

Cui & Zhang (2024) have ascertained that users combining bikes with bus and rail trips reported improved urban transport, emphasizing key benefits including reduction in travel times, easy accessibility, and reliable urban transport alternatives. Factors that have contributed to the usage of the Blue Bike system in Boston and sister cities are proximity to transit stations (NACTO 2023), Weather and Seasonality (Fishman, 2015), and pricing and Equity Programs (MACP 2023). Integration with Transit (Ethier et al., 2024) and Technology (Shaheen, 2019). The empirical evidence reported by prior studies has proven the contributing role of the Blue Bike system in enhancing transit connectivity and multimodal transport. However, the contribution to greenhouse gas emissions is not explicitly explored.

3.0 Research Methodology

3.1 Research Design and Search Strategy

As mentioned by Kitchen (2004), cited in P.A Anokye et al (20--), one of the critical steps in conducting scientific research is systematic review. The approach involves critical analysis of already published literature which deem relevant to the research topic, research questions, and objectives under studying through identification, mapping, aggregating, evaluating, and synthesizing them. Again, it also helps to identify a research gap. The study employed a Systematic Review approach using the Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) approach present a broader view of the research objectives and questions by synthesizing relevant literature, ensuring little to no bias, which aligns with accepted systematic review procedure. For this study, to answer the research questions, the authors conducted a systematic literature search using the advanced search from ScienceDirect to fetch literature from credible scientific journals. The authors prioritize literature published between the years 2010 and 2026 purposely to narrow down the search and content, and also to report and rely on current information that presents current happenings regarding the blue bike system and its effect on greenhouse gas emission mitigation. For easy accessibility of peer-reviewed articles and other relevant documents, the research focused on ScienceDirect as the Database to source these relevant papers.

ScienceDirect was selected as a result of its wider scope of coverage of high-quality peer-reviewed research articles spanning multiple disciplines to ensure a robust and credible justification for the study. Since the aim of the study is set to a) assess spatial linkage of Blue Bike stations and MBTA transit nodes, b) examining the extent of automobile trip substitutes regarding trip characteristics, usage intensity, and threshold and c) lastly assessing greenhouse gas (CO₂) emission reduction as a result of the Blue Bike initiative, the research used different Boolean search approaches and keywords at different stage of the search. For instance, keywords such as *Bluebikes*, *bikeshare*, *bike sharing*, *bicycle sharing*, *transit connectivity*, *modal integration*, *first-mile*, *last-mile*, *accessibility*, *greenhouse gas emissions*, *GHG emissions*, *carbon emissions*, *emission reduction*, and *sustainability* were used to conduct the

literature search. The main Boolean operators used were “AND,” “OR,” and “LIMIT-TO,” see **Table 1**. Each of these operators was used for a specific function or role in the literature search. For example, the operator “AND” was used to integrate the keywords in the phrase, “OR” was used to join identical or similar keywords, and the operator “LIMIT-TO” was used to narrow the search to only the Boston Metropolitan Area, papers published between the years 2020 and 2026, and papers published in the English Language. The primary stage of search using the above keywords mentioned yielded 509 documents related to the topic. Out of the 509 documents, 457 were peer-reviewed articles, while fifty-two of them appeared as grey literature, and this was to capture a broader range of studies related to the topic.

Table 1: Literature search stages and search strings used.

Search stages	Search String
Stage 1: To acquire broader range of studies	(Bluebikes OR bikeshare OR "bike sharing" OR "bicycle sharing") AND ("transit connectivity" OR "modal integration" OR "first-mile" OR "last-mile" OR accessibility) AND ("greenhouse gas emissions" OR "GHG emissions" OR "carbon emissions" OR "emission reduction" OR sustainability)
Stage 2: Narrowing content to Boston Metropolitan Area	(Bluebikes OR bikeshare OR "bike sharing" OR "bicycle sharing") AND ("transit connectivity" OR "modal integration" OR "first-mile" OR "last-mile" OR accessibility) AND ("greenhouse gas emissions" OR "GHG emissions" OR "carbon emissions" OR "emission reduction" OR sustainability) LIMIT-TO (Boston)

Source: Produced by Authors (2026).

3.2. Selection of relevant literature using inclusion and exclusion criteria.

To narrow the content to Boston, a second search was conducted using a revised search string (see Table 1). Based on the second search strings and other defined inclusion and exclusion criteria (see Table 2) set by the authors, the 509 documents were subjected to screening. A total of 146 duplicates were removed, while 363 articles were screened based on title and abstract. The screen was conducted in line with the objectives of the research, where priority was given to documents that were highly relevant and had the data needed for analysis. Out of 363 documents screened, 138 full-text articles were assessed.

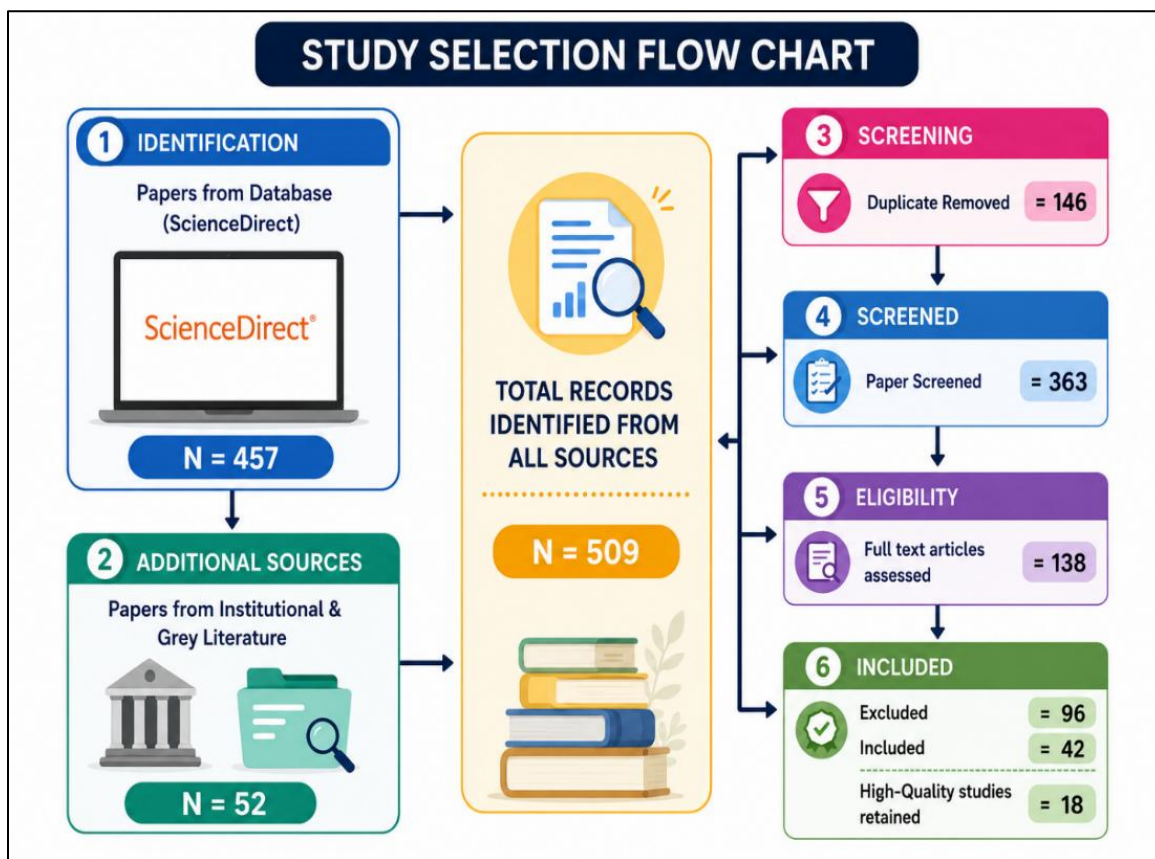
Applying the inclusion and exclusion criteria set in Table 2, 96 out of the 138 documents were excluded, while forty-two were included. As demonstrated in the flow chart diagram, after applying the Joana Briggs Institute Checklist to conduct quality appraisal and check for eligibility, the number of excluded documents reduced from 42 to 18 as eligible literature to answer the research questions and objectives. These processes were done manually by the authors without using any software as a supporting tool.

Table 2: Exclusion and Inclusion criteria.

Focused Areas	Exclusion criteria	Inclusion criteria
Language	<i>Any article or document written in different language other than English.</i>	<i>Only articles or document written in English</i>
Document type	<i>Blogs, social media post, magazines, and articles without rigor scientific methodology</i>	<i>Peer-reviewed journal article, book chapters, grey literature with good and accepted scientific methodology</i>
Content focused	<i>Research works that did not discuss blue bike or bikeshare systems, multimodal transport systems, spatial linkage of blue bike or bikeshare stations, GHG</i>	<i>Research works that discuss blue bike or bikeshare systems, multimodal transport systems, spatial linkage of blue bike or bikeshare stations, GHG</i>

Methodology	impacts, and automobile trip substitution Studies that did employed either qualitative or quantitative approach or both	impacts, and automobile trip substitution. Studies that employed either qualitative or quantitative approach or both
Publication year	Studies published before 2010.	Studies published between 2010 and 2026
Geographical scope	Studies conducted outside Boston Metropolitan Area	Studies conducted within Boston Metropolitan Area

Source: Produced by Authors (2026).



Source: Produced by Authors (2026)

4.0 Analysis and Result

4.1 Spatial Linkage of Blue Bike Stations and Boston (MBTA) Transit Node

Assessing the spatial linkage of Blue Bike Stations and the connectivity to various transit nodes, specifically rail and bus stop points, empirical evidence from the multiple selected studies revealed that in Boston, the Blue Bike system connects more than 400 public transit pickup and stop points (Ding et al., 2025; Shaheen, Susan, 2019). As reported by Ding et al (2025), the Blue Bike network is deliberately co-located in proximity to the Massachusetts Bay Transportation Authority (MBTA) subway, bus, and commuter rail networks, facilitating first and last-mile trip connectivity to areas such as Boston, Brookline, Somerville, and Cambridge. Ding et al (2025) employed a GIS application and Blue Bike trip data to model the spatial correlation of Blue Bike station connectivity to MBTA transit nodes between the period of 2015 and 2024. The findings from the study revealed that most of Blue Bike stations across Boston are located within a 400-meter buffer zone, providing a robust and enhanced system for last-mile accessibility.

Findings from Bergman & Allenspach (2025) and Joyce-Johnson (2025) revealed that Blue Bike stations close to dense transit hubs, especially those cited close to South Station, Back Bay, and Kendall Square, resulted in an increase in the volume of trips of about 2.5 times higher than in peripheral areas. This point underscores the spatial interaction between rail infrastructure and the Bike share or Blue Bike. This is aligned with evidence presented in cities such as New York City and Chicago, where trip densities were found to peak within a short walking distance around subway nodes.

Considering the Blue Bike network configuration, it was demonstrated by Volker et al (2020) that the high intramodality of Blue Bike overlapping with catchment areas of transit nodes increased the frequency and system quality, enhancing ridership. Again, within the context of the broader Greater Boston Metropolitan Area, there is high correspondence to higher trip intensity, with about 4.2 mean per ride per day within areas with integrated Bikeshare coverage, while areas within 800 meters and above from a transit hub witnessed a mean ride per day of 1.6 (Volker et al., 2020). Not only the increase in intensity, but also a study

conducted by Rush et al (2022) using aggregated shared mobility data revealed that the location and spatial distribution of stations contributed to the growth of bikeshare ridership, with an approximate variance of 35% between the period of 2017 and 2022. Though the Blue Bike system has enhanced ease of mobility and strengthened multimodal trips across the Boston Metropolitan Area, empirical evidence from Krauss et al (2022) and Braunlin (2023) has affirmed spatial inequality. Evidence from these studies revealed that, while the system has been efficient, available, and dependable within all parts of the Boston Metropolitan Area, especially high-density areas in central Boston, areas within the outskirts, for example, Hyde Park and Mattapan, remained underserved.

Table 3: Extracted Data, Summary of evidence and analysis from literature.

Author	Study Area	Method used	Spatial Linkage Variables	Key findings relevant to the objective	Degree of relevance
(Ding et al., 2025)	Exclusively on Boston	Longitudinal spatial analysis	Proximity of bike stations to transit, station distribution	Blue Bike stations are strategically located near public transport corridors, improving multimodal accessibility and facilitating transit integration.	High
Joyce-Johnson (2025)	Chicago but included Boston as part of the analysis	Mixed methods	Shared micromobility, transit interaction, first-and-last mile	Shared micromobility strengthens first- and last-mile connections and enables car-free travel through transit integration	High

Boeri (2025)	New York City but included Boston as part of the analysis	User behaviour analysis	Protected bicycle facility and network connectivity	Protected cycling infrastructure improves accessibility between micromobility systems and transit facilities	Moderate
Braunlin (2023)	Metropolitan Cities but included Boston as part of the analysis	Policy review	Equity and infrastructure distribution	Unequal spatial distribution of micromobility infrastructure affects accessibility and transport equity. Data show that shared mobility systems are concentrated around urban activity centres and transit hubs	Moderate
Rush et al (2022)	United States but included Boston as part of the analysis	Data review	Shared mobility data and spatial availability	Bike-share systems are most effective when integrated with public transit and located near high-demand destinations	High
Shaheen Susan (2019)	Global but included Boston as part of the analysis	Policy toolkit	Dock location strategies	Policy interventions increased utilization in areas with better transit access	High
Smith, (2022)	United States but included Boston as part of the analysis	Evaluation study	Station accessibility		Moderate

Zou (2023)	Multiple cities including Boston	Evaluation study	Sustainable mobility integration	Micromobility complements public transport by extending service coverage and reducing accessibility gaps. Strategic station placement around commercial and transit areas enhances network efficiency and accessibility	High
Bergman & Allenspach, (2024)	Chicago but included Boston as part of the analysis	Case study	Bicycle station distribution		High

Source: Produced by Authors (2026).

4.2 Behavioural Characteristics and Automobile Trip Substitution by Blue Bike in the Greater Boston Metropolitan Area

The empirical data show that about 11% of travellers directly replace vehicular (car) trips with Bike Share, especially in short-distance trips of about 5km and below (Ding et al., 2025). This trip substitution is effective and more pronounced during weekday peak hours, where bike share use aligns with commuting patterns. Findings from He et al (2020) show that there is a positive correlation between the fluctuation in prices of gasoline and bikeshare ridership. The author ascertained that 10% increase in gasoline prices also leads to about 12% patronage and usage of Blue Bikes trips, demonstrating price-sensitivity and its behavioural influence on travellers. The post-pandemic era also witnessed a shift from more vehicular trips to bikeshare (Smith, 2022). For example, during the COVID-19 pandemic era and post-COVID-19, an 80% rebound in Blue Bike ridership was recorded for short trips, while the usage of vehicles declined by 14% within the inner parts of the Boston Metropolitan Area. About a 17% increase

in shared micromobility (Blue Bike) at the expense of vehicular trips in the year 2019, where urban dwellers and city residents sought for low-contact and flexible mode of transport. In terms of Blue Bike trip characteristics, it was found that the average trip length was approximately 2.1km, corresponding to a time duration of 13 to 15 minutes ride (Volker et al., 2020). These trips were found to be clustered around commercial and dense employment centres as a substitute for vehicular commuters and a complement to short-distance transit trips.

It was also found that the intensity of Blue Bike usage (trips per bike per day) increased from 1.8 to 3.6 on average from the period of 2018 to 2023 (Ding et al., 2025) indicating a growing interest and the adoption of the Blue Bike system. According to Sanchez et al (2022) and Kou et al (2020b), users often substitute vehicular trips with bikeshare, specifically when assisted by improved and safe infrastructure and stations for cycling proximity to origin-destination points. The perception that cycling lanes are highly safe also influences the substitution of bike sharing for vehicular trips. Regarding behavioural segmentation, it was found that there are demographic disparities in the usage of Blue Bike during peak hours. For instance, it was found that compared to women of the same age group, men between 25 and 44 were more likely to often substitute a car trip for a bikeshare trip during peak hours. This, therefore, indicates that women and older men showed a lower substitution rate (Braunlin, 2023; Shaheen Susan, 2019).

4.3 Contribution of Blue Bike to Greenhouse Gas Emission within Boston

This objective examines the contribution of the Blue Bike system to Greenhouse Gas (GHG) emissions across the Boston Metropolitan Area. The result and empirical evidence of all the documents reviewed showed that the Blue Bike system has played a crucial role in CO₂ emissions by a measurable quantity through the direct replacement of car trips by bikeshare or ridership. Findings from Kou et al (2020) ascertained that the Blue Bike system has yielded an impact on greenhouse gas emissions with an estimated reduction of about 0.5kg to 1.2kg per trip, primarily as a result of a shift from car trips to bike sharing, especially within central parts of Boston, Cambridge, and Brookline, with improved infrastructure. However, the

findings from Ding et al. (2025) have highlighted the recent status of greenhouse gas emissions reduction, which aligns with Kou et al (2020). Ding et al (2025) employed longitudinal analysis and a dataset of 20.07 million bike trips within Boston from the period of January 2015 to December 2024. According to findings from the study, the Blue Bike initiative has contributed to a net reduction of 7614.96 tons of CO₂ as a result of the shift from vehicular (car) trips to bike-share usage. The author employed multiple approaches, including the integration of Scheiner's model and the Multinomial Logit model, and optimized the Seasonal Autoregressive Integrated Moving Average. While substitution of blue bike trips was assessed using Scheiner's model and Multinomial Logit model, usage and energy saving (emission reduction) were calculated using the optimized Seasonal Autoregressive Integrated Moving Average.

The study affirmed a corresponding reduction rate of 0.4% within the Greater Boston Metropolitan Area. Krauss et al (2022) also found that shared micromobility has collectively contributed to greenhouse gas emissions by 0.8% to 2.1% within Boston. Findings from Bergman & Allenspach (2024) show a net reduction ratio in greenhouse gas emissions of 3.5:1 comparing avoided emissions to operation emissions. A meta-analysis conducted by Steckler et al. (2022), which reviewed data from over 40 shared mobility studies, confirmed that shared micromobility reduces urban CO₂ emissions by 4% to 8%, with even greater effects observed in densely populated, multimodal cities in the context of Boston. Blue Bikes' system is notable for its contribution to GHG mitigation, attributed to its integration with the MBTA network and a high daily ridership exceeding 35,000 trips as of 2024.

4.5 Discussion of Results.

Existing literature has documented key insights into bike-share accessibility, user behaviour, and the spatial equity dimension of the system within Boston. However, the fragmented evidence on the spatial linkage of bike-share stations and transport nodes, extent of trip substitution, trip characteristics, intensity of usage, and synthesis of empirical evidence on the contribution of blue bikes to greenhouse gas mitigation in Boston formed the core argument and justification of this study. Therefore, the study sought to 1) assess the spatial linkage between Blue Bike stations and MBTA transit nodes, 2) examine the extent of vehicular (car) trip substitution, trip characteristics, and usage intensity, and lastly, Blue Bike's contribution to greenhouse gas emission reduction.

With regard to spatial linkage between blue bike stations and MBTA transit nodes, the findings from the review asserted that, in the context of the Boston Metropolitan Area, blue bike is not just a mere cycling system; it is strategically integrated in the city's urban transport models to complement MBTA networks and also foster a multimodal transport system within the enclaves of Boston. According to findings from the review, many of the blue bike stations within Boston are situated not longer than four hundred meters around a transit node. This was found to be a strategic spatial location of these systems to deliberately provide a robust and enhanced system that will promote last-mile accessibility (Ding et al., 2025). These findings are affirmed by studies like (Bergman & Allenspach 2024; Rush et al., 2022; Shaheen Susan, 2019), which asserted that the proximity of bike-share infrastructure to transit nodes has increased the fostering of multimodal travel in Boston. On the other hand, the revealed spatial equity in terms of infrastructure provision. For instance, Findings from (Braunlin, 2023; Krauss et al., 2022) highlighted that despite the level of integration of the system in Boston's urban transport modalities, there is a high level of concentration of the system, particularly within central Boston, while the peripheral area remains underserved. The implication of these findings is that the effectiveness of the Blue Bike is attributed to both the spatial distribution of stations and integration, and not solely to station density.

For trip substitution, characteristics and user behaviour, findings from the review provide key insight into the behavioural role of Blue Bikes within Boston's urban transport system. Evidence from the literature revealed that the system has been a substitute for shorter distances and trips, which at first were primarily car-dominant, especially trips within 5km and during rush hours. However, not all forms of motorised trips (Ding et al., 2025; Volker et al., 2020). This aligns with findings from Joyce-Johnson (2025), which affirmed blue bike as an effective first-and-last-mile system, reducing over-reliance on private cars. In addition, it was found that trip substitution is primarily influenced by contextual factors such as fuel prices, cycling infrastructure, and proximity to destination (He et al., 2020; Kou et al., 2020b; Sanchez et al., 2022). Therefore, it is noted that within the context of Boston, blue bike systems do not universally replace automobile (vehicular) trips; however, it occurs under favourable urban conditions, routine commuting, and local mobility. This gives a nuanced interpretation which extends and advance exiting body of knowledge by highlighting that travel behaviour is subject to conditions such as infrastructure and urban form but not being an inherent outcome of bike-share availability.

Moreover, the findings highlighted different user and trip characteristics that influence blue bike utilization and patronage within Boston. The review found that primarily dominant users of the system are working-age commuters embarking on short trips, particularly with employment centers and transit corridors (Braunlin, 2023; Volker et al., 2020). Again, the intensity of the usage of the system was found to have increased steadily between the period of 2018 and 2024, reflecting the emerging interest and adoption of the blue bike as an alternative urban transport mode (Ding et al., 2025). However, this growth was found to be unevenly distributed across locations and distinct groups. For instance, the findings from Ding et al. (2025) reported a low substitution rate of usage among women and the aged. This implies that there are behavioural and infrastructure barriers that influence patronage or usage of the system among these user groups.

With regard to greenhouse gas emission reduction, the synthesis evidence from the literature affirmed some level of contribution made by the blue bike system to mitigate emission

reduction within the enclaves of the Boston Metropolitan Area. It was found that the blue bike system has contributed to emission reduction through the replacement of car trips within short distances, particularly distances below 5 km. However, the evidence found in the literature supports associative interpretation rather than causal. For instance, Ding et al (2025) found that an increase in blue bike usage leads to a measurable reduction in carbon emissions. Specifically, the author reported an estimated net reduction of 7614.96 tons of CO₂ as a result of the shift from vehicular (car) to bike-share usage. This aligns with findings from studies such as (Krauss et al., 2022; Steckler et al., 2022), which also found that shared micromobility has contributed to low-transport-related emissions within Boston, particularly where this multimodal transport exists. However, because this study employed a meta-synthesis systematic review and not primary modelling or experimental analysis, this study therefore concludes that blue bike alone causes observed emission reduction, rather it demonstrates that blue bike systems have played a contributing role in greenhouse gas emission reduction, especially in areas where it has successfully substituted car trips and also operated within an integrated urban transport network.

The key contribution to literature or additions of knowledge of this study lies in its integrated synthesis of evidence across spatial linkages between blue bike stations and transit nodes, travel behaviour, and environmental outcomes of the blue bike system within the context of the Boston Metropolitan Area. While existing literature examined these dimensions independently, leading to fragmented evidence on the subject matter, this study has provided a synthesis of evidence on the spatial linkage, behavioural characteristics, trip substitution, and usage intensity, as well as addressing the critical question of the contribution of the blue bike to greenhouse gas emission within Boston. For instance, this study has been able to consolidate fragmented evidence in the literature by providing a comprehensive understanding of how Blue Bikes functions within a multimodal urban transport ecosystem within Boston while carefully avoiding unsupported causal claims. This offers nuanced evidence of blue bike contributions to urban transport and environmental sustainability. Again, this study has extended and added an in-depth understanding to existing literature by

linking blue bike usage intensity to user groups and infrastructural characteristics at the expense of treating ridership growth as an isolated outcome.

4.6 Limitations of the study.

This study acknowledges several limitations that need to be considered when interpreting the study's findings. To start with, the study used a systematic review and meta-synthesis approach, which means it relied solely on secondary data and evidence from previous studies. As a result, the study's findings are entirely dependent on the methodological approach of the studies reviewed, quality, and scope, and hence could not be independently verified through primary data collection. Again, the availability of Boston-specific empirical evidence and studies was limited, leading to the inclusion of evidence from comparable city contexts and analysis where Boston was integrated to support the analysis. While these studies offered key insights, this study acknowledges that contextual differences may influence the applicability of some findings in the context of Boston.

Finally, this study did not conduct primary modelling or causal analysis; rather, it synthesized the association and key findings reported in the literature. As a result, findings from the study are considered indicative of potential contributions and relationships instead of causal effects between blue bike, transit connectivity, and greenhouse gas emission reduction within the context of the Boston Metropolitan Area. Therefore, future research should focus on providing empirical evidence on the relationship between bike-share, transit connectivity, and travel behaviour using primary data and advanced spatial modelling techniques. Also, future studies can consider examining the equity and accessibility implications of the Blue Bike across diverse user groups to determine inclusiveness in the design of the system and its influence on usage intensity.

5.0 Conclusion.

This study aimed at synthesising existing evidence on the contribution of the blue bike to transit connectivity and greenhouse gas emission reduction within the context of the Boston Metropolitan Area. The study specifically examined the spatial linkage between blue bike stations and MBTA transit nodes, the extent of automobile (car) trip substitution together with user and trip characteristics, and the contribution of blue bikes to greenhouse gas emission reduction. Employing a systematic review and meta-synthesis approach, this study addressed fragmented evidence regarding the contributions made by blue bikes to urban transport and environmental sustainability.

The study's findings indicate that within the context of the Boston Metropolitan Area, Blue Bikes is integrated with MBTA nodes and serves as first-and-last -mile mobility. The findings also highlighted that the effectiveness of this integration can not only be attributed to the proximity of stations to transit nodes; however, the spatial distribution of this infrastructure across neighbourhoods also played a crucial role. Also, it was found that blue bike substitutes short car trips and journeys less than 5 km, particularly with those associated with routine and local mobility. However, trip substitution is found to be influenced by contextual factors such as urban form, accessibility, and availability of infrastructure, instead of occurring uniformly across all travel types. The findings also indicated that usage of the system is concentrated among the working-age commuters embarking on short trips, and usage intensity differs across locations and groups. Again, the study's findings affirmed that the blue bike has potentially contributed to the reduction of greenhouse gas emissions within the enclaves of the Boston Metropolitan Area.

However, this evidence supports an associative contribution at the expense of a direct causal relationship between blue bikes and emission reduction outcomes. By synthesising fragmented evidence on spatial linkage, travel behaviour, and contributions to greenhouse gas emission, this study has provided a more holistic understanding of how the Blue Bike has contributed to the Boston multimodal urban transport system and has provided consolidated

evidence for sustainable urban mobility policy design, transport planning, and evidence for future research focus.

DECLARATIONS

Ethics of Approval and Consent to Participate - Ethics and Consent to Participate declaration: not applicable.

Consent to Publish - Consent to Publish declaration: not applicable.

Availability of Data and Materials

Not applicable.

Competing Interest - The authors declare that they have no known financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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AUTHORS CONTRIBUTIONS

Donkor Ankrah contributed to the study design, data collection, data analysis, and manuscript revision.

Moses Ayirebi conceptualized the study, conducted the literature review, analyzed the data, and drafted the original manuscript.

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